

TECHNICAL MEMORANDUM

TO: Sage Creek Irrigation District — Board of Directors

FROM: M. L. Brennan, P.E., and T. R. Vasquez, P.E. High Desert Engineering, LLC

DATE: April 15, 2025

RE: Canal Lining Design Alternatives — Reach 7 Rehabilitation Sage Creek Main Canal, Sta. 142+00 to 166+00

PROJECT: HDE No. 2025-IR-0012

1.0 PURPOSE

This memorandum evaluates four canal lining alternatives for Reach 7 of the Sage Creek Main Canal based on the geotechnical investigation findings (HDE, March 2025). The evaluation considers performance against the two primary subgrade challenges — expansive lacustrine clay (swell pressure 3,600 to 6,500 psf) and collapsible loess — as well as seepage control, durability, constructibility, and lifecycle cost.

2.0 DESIGN CRITERIA

Table 2-1: Lining Design Criteria

Parameter	Requirement	Basis
Design flow	280 cfs (full capacity)	SCID operations
Canal geometry	14-ft bottom, 2H:1V left / 3H:1V right bank	Geotech recommendation
Lining height	7.0 ft (5.5 ft flow + 1.5 ft freeboard)	USBR standard
Seepage target	< 1.0 cfs per 1,000 ft	USBR lined canal criteria
Swell resistance	Withstand 3,600-6,500 psf uplift	Lab testing (ASTM D4546)
Freeze-thaw	300+ cycles design life	ASTM C666 (Magic Valley)
Design life	40 years minimum	USBR / WaterSMART requirement
Velocity	Max 8 ft/s (Manning's n = 0.013-0.015)	USBR EM-7

3.0 LINING ALTERNATIVES

3.1 Alternative A: Reinforced Concrete with Geomembrane (Recommended)

Composite lining: 40-mil HDPE geomembrane between granular cushion layers, with 4-inch reinforced concrete overlay. Lime-treated subgrade beneath.

Cross-section (bottom to top):

- Prepared subgrade: lime-treated clay (6% CaO, 8-in depth, 95% Std Proctor, 7-day cure)

- Lower cushion: 3-inch compacted sand bedding (ASTM C33)
- Geomembrane: 40-mil smooth HDPE, thermally welded seams, 100% tested
- Upper cushion: 3-inch compacted sand cover
- Concrete lining: 4-inch reinforced (4,000 psi, 6x6-W2.9xW2.9 WWF, 6±1% air)
- Joints: transverse contraction at 12 ft, PVC waterstop, polyurethane sealant

Advantages: Highest seepage reduction (99.7%); geomembrane isolates concrete from subgrade moisture, reducing swell and freeze-thaw; reinforcement controls cracking; proven on USBR canals in the region (Milner-Gooding, North Side). **Disadvantages:** Highest initial cost; requires careful membrane QA.

3.2 Alternative B: Geomembrane Only (Exposed)

60-mil textured HDPE geomembrane placed on lime-treated subgrade with no concrete overlay. **Advantages:** Lowest initial cost; excellent seepage control (95%+); fast installation; flexible, tolerates subgrade movement. **Disadvantages:** UV degradation limits life to 10-15 years; susceptible to mechanical damage; higher Manning's n (0.025) reduces capacity; poor ice resistance; does not meet 40-year life for WaterSMART eligibility.

3.3 Alternative C: Fiber-Reinforced Shotcrete (4-inch)

Pneumatically applied concrete with polypropylene fiber reinforcement over lime-treated subgrade, no geomembrane. **Advantages:** Good structural performance; conforms to irregular subgrade; rapid application; proven for canal rehabilitation. **Disadvantages:** No moisture barrier — seepage reduction limited to 90-95%; subgrade moisture can still drive swell from below; surface roughness ($n = 0.016-0.018$); 15-20% material loss from rebound.

3.4 Alternative D: Unreinforced PCC (Replace in Kind)

New unreinforced 4-inch concrete panels with improved mix (non-reactive aggregate, air-entrainment, low w/c). Same panel dimensions as original 1962 installation. **Advantages:** Lowest technical complexity; familiar to contractors. **Disadvantages:** Repeats the design that failed — without a moisture barrier, the panels face the same 3,600-6,500 psf swell forces. Estimated life: 20-25 years. **Not recommended.**

4.0 ALTERNATIVES COMPARISON

Table 4-1: Lining Alternatives Comparison Matrix

Criterion	Alt A RC + Membrane (RECOMMENDED)	Alt B Membrane Only	Alt C Shotcrete	Alt D Unreinforced PCC
Seepage reduction	99.7%	95%+	90-95%	90-95%
Swell resistance	Excellent	Good	Fair	Poor
Freeze-thaw	Excellent	N/A	Good	Good
Design life	40-50 yr	10-15 yr	30-35 yr	20-25 yr
Manning's n	0.013-0.015	0.025	0.016-0.018	0.013-0.015
Project cost	\$1,020,000	\$520,000	\$780,000	\$720,000
40-yr lifecycle (NPV)	\$1,180,000	\$1,620,000	\$1,340,000	\$1,580,000
WaterSMART eligible	Yes	No	Yes	Yes
40-yr seepage savings	\$5,400,000	\$4,800,000	\$4,500,000	\$4,500,000
Net benefit (40 yr)	\$4,220,000	\$3,180,000	\$3,160,000	\$2,920,000

Alternative A provides the highest 40-year net benefit (\$4.22M), longest design life, best seepage control, and best swell resistance. While it has the highest initial cost, the lifecycle cost is lowest because it requires no mid-life replacement.

5.0 RECOMMENDED LINING SPECIFICATION

Table 5-1: Lining System Specification Summary

Component	Specification	Standard	Notes
Lime treatment	6% CaO, 8-in depth, 95% Std Proctor, 7-day cure	ASTM D6276 ASTM D698	Verify PI < 15
Sand cushion (×2)	3 in each, ASTM C33 concrete sand	ASTM C33	Protect membrane
Geomembrane	40-mil smooth HDPE, thermal fusion weld, 100% seam testing	ASTM D4439 GRI GM-13	Min 6-in overlap
Concrete	4,000 psi, Type II LA, w/c ≤ 0.45, 6±1% air	ASTM C150 ASTM C260	Non-reactive agg (ASTM C1260)
Reinforcement	6x6-W2.9xW2.9 WWF, 1.5 in clear cover	ASTM A185	Place on chairs
Joints	12-ft transverse, PVC waterstop, polyurethane sealant	ASTM D2240	Sawcut ¼ depth within 24 hr
Drainage blanket (right bank)	12 in crushed gravel, 4-in perf HDPE at toe	ASTM D2321	Discharge to canal downstream

6.0 CONSTRUCTION SEQUENCE

- 1. Demolition:** Remove existing lining and debris. Stockpile salvageable concrete for crushing (potential use as granular fill on other SCID projects).
- 2. Right bank re-grading:** Cut bank from 2H:1V to 3H:1V. Install drainage blanket (12-inch gravel) and perforated drain pipe at bank toe. Export approximately 2,400 CY.
- 3. Subgrade preparation:** Scarify upper 8 inches of exposed canal subgrade. Spread hydrated lime at 6% by dry weight using a spreader truck. Mix with rotary mixer to full 8-inch depth. Compact to 95% Standard Proctor. Allow 7-day minimum cure. Verify PI reduction to below 15 by field sampling and Atterberg testing.
- 4. Lower sand cushion:** Place and hand-compact 3 inches of ASTM C33 concrete sand over the lime-treated subgrade. Smooth to uniform grade.
- 5. Geomembrane:** Deploy 40-mil HDPE panels from invert to bank crest with 6-inch minimum overlap at seams. Thermally weld all seams. Perform destructive and non-destructive seam testing per GRI GM-19. Anchor at bank crests in 12-inch-deep anchor trenches.
- 6. Upper sand cushion:** Place 3 inches of sand cover over the membrane. Avoid construction equipment directly on the membrane before sand placement.
- 7. Concrete placement:** Place 4-inch reinforced concrete using a canal paver or slipform equipment. Maintain 1.5-inch cover over WWF. Sawcut transverse joints at 12-foot spacing within 24 hours of placement. Install PVC waterstop at all joints prior to concrete placement. Apply curing compound immediately.
- 8. Joint sealing:** After 28-day cure, clean joints and apply polyurethane sealant.
- 9. Commissioning:** Fill canal slowly (6 inches per hour) and monitor for leaks. Record initial seepage measurements at flumes for comparison with pre-rehabilitation values.

7.0 COST ESTIMATE AND SCHEDULE

Table 7-1: Detailed Cost Estimate — Alternative A

Item	Quantity	Unit Cost	Total
Demolition / removal	28,800 SF	\$3.50/SF	\$101,000
Right bank re-grading	2,400 CY	\$18/CY	\$43,000
Drainage blanket + pipe	4,300 SF	\$8/SF	\$34,000
Lime treatment	2,140 CY	\$22/CY	\$47,000
Sand cushion (both layers)	1,800 CY	\$28/CY	\$50,000
Geomembrane (40-mil HDPE)	32,500 SF	\$3.50/SF	\$114,000
Reinforced concrete lining	28,800 SF	\$14/SF	\$403,000
Joints and waterstop	2,400 LF	\$12/LF	\$29,000
Mobilization (8%)	LS	—	\$66,000
Contingency (15%)	LS	—	\$133,000
TOTAL			\$1,020,000

Schedule: Construction during the non-irrigation season (October through March). Estimated duration: 10 to 12 weeks. Critical path: lime treatment cure time (7 days minimum) and concrete cure time (7 days before joint sawcutting). The project can be completed in a single off-season if awarded by August 2025.

Funding: SCID should apply for USBR WaterSMART Water and Energy Efficiency Grant (up to 50% federal cost-share, \$500,000 maximum). Applications are due annually in October. The 5,400 AF/yr seepage savings and 40-year design life make this project highly competitive for WaterSMART funding. Simple payback: 3.8 years at current water rates (\$50/AF).

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